

Project Design Phase

Solution Architecture

DATE	TEAM ID	PROJECT NAME	MAXIMUM MARKS
03 NOV 2025	NM2025TMID07720	CRM Application for Jewel Management(Developer)	4 Marks

Goals of the Architecture:

- Provide a unified system to manage customer data, jewelry stock, and billing efficiently.
- Ensure data integrity and reduce manual work through process automation.
- Enhance communication between customers and jewelry staff via automated notifications.
- Offer real-time insights and analytics for sales and inventory management.

Key Components:

- Customer Database (Stores customer profiles, purchase history, and preferences).
- Jewelry Inventory Module (Tracks stock levels, item categories, and availability).
- Order & Billing System (Handles order placement, invoicing, and payment tracking).
- Automation Flows (Automates order updates, billing, and notifications).
- Reporting Dashboard (Provides analytics on sales, stock, and customer behavior).
- Salesforce Cloud Platform (Centralized hosting and data security).

Development Phases:

1. Design the data model for customers, inventory, and orders in Salesforce.
2. Create automation workflows for order processing and billing.
3. Develop Apex Classes for business logic and validation rules.
4. Integrate payment gateway APIs and notification services.
5. Test each module for data accuracy, speed, and reliability.

Solution Architecture Description:

The CRM Application for Jewel Management is designed to provide an end-to-end digital solution for jewelry stores. It connects various operational areas—customer engagement, stock management, order processing, and billing—through a centralized Salesforce

platform. The architecture ensures seamless data flow between components, improving efficiency and reducing manual interventions.

Each module within the CRM communicates via Salesforce's cloud-based database, allowing real-time updates and synchronization. When a customer places an order, the system automatically updates inventory, generates an invoice, and triggers a notification. Managers can monitor all operations using dashboards and analytics reports, enabling better decision-making.

This architecture minimizes human error, improves sales tracking, and maintains a smooth flow of operations. It is scalable for multi-branch jewelry businesses and can easily integrate with additional modules such as AI-based recommendation systems and loyalty programs.

Reference:

<https://www.salesforce.com/products/platform/overview/>

<https://trailhead.salesforce.com/>

<https://aws.amazon.com/blogs/industries/architecture-and-design-considerations-for-cloud-based-crm-solutions/>